3.1.3 The halogens

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Characteristic physical properties		
(a)	existence of halogens as diatomic molecules and explanation of the trend in the boiling points of Cl_2, Br_2 and I_2, in terms of induced dipole–dipole interactions (London forces) **(see also 2.2.2 k)**	
Redox reactions and reactivity of halogens and their compounds		
(b)	the outer shell s^2p^5 electron configuration and the gaining of one electron in many redox reactions to form 1– ions	Throughout this section, explanations of redox reactions should emphasise electron transfer and oxidation number changes and include full and ionic equations **(see also 2.1.5 Redox)**.
(c)	the trend in reactivity of the halogens Cl_2, Br_2 and I_2, illustrated by reaction with other halide ions	Including colour change in aqueous and organic solutions.
(d)	explanation of the trend in reactivity shown in **(c)**, from the decreasing ease of forming 1– ions, in terms of attraction, atomic radius and electron shielding	
(e)	explanation of the term *disproportionation* as oxidation and reduction of the same element, illustrated by: **(i)** the reaction of chlorine with water as used in water treatment (ii) the reaction of chlorine with cold, dilute aqueous sodium hydroxide, as used to form bleach (iii) reactions analogous to those specified in **(i)** and **(ii)**	
(f)	the benefits of chlorine use in water treatment (killing bacteria) contrasted with associated risks (e.g. hazards of toxic chlorine gas and possible risks from formation of chlorinated hydrocarbons)	HSW9,10,12 Decisions on whether or not to chlorinate water depend on balance of benefits and risks, and ethical considerations of people's right to choose. Consideration of other methods of purifying drinking water.
Characteristic reactions of halide ions		
(g)	the precipitation reactions, including ionic equations, of the aqueous anions Cl^-, Br^- and I^- with aqueous silver ions, followed by aqueous ammonia, and their use as a test for different halide ions.	Complexes with ammonia are **not** required other than observations. **PAG4 (see also 3.1.4 a)** HSW4 Qualitative analysis.